

3.2: Quadratic functions

Definition.

A quadratic is a function of degree two. The **general form of a quadratic function** is

$$f(x) = ax^2 + bx + c \quad (a \neq 0)$$

where a , b , and c represent constants. The **standard form of a quadratic** is

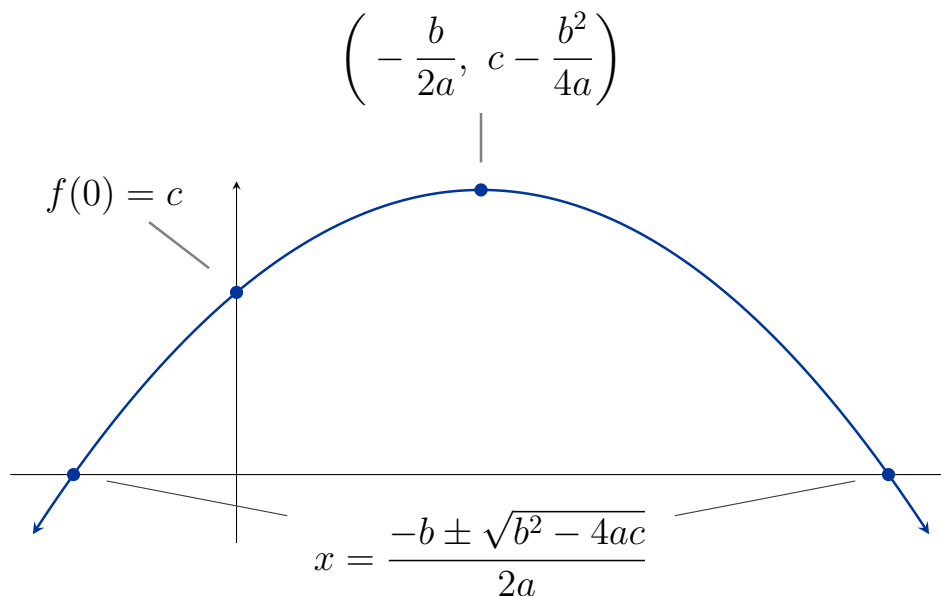
$$f(x) = a(x - h)^2 + k.$$

A **parabola** is the shape of the graph of a quadratic function, with a vertex at (h, k) :

$$h = -\frac{b}{2a}, \quad \text{and} \quad k = f(h) = c - \frac{b^2}{4a}$$

The optimal value occurs at the vertex of a parabola:

- A maximum if $a < 0$ 
- A minimum if $a > 0$ 



Example. Expand the standard form of a quadratic, then write h and k in terms of a , b , and c .

$$\begin{aligned} a(x-h)^2 + k &= a(x^2 - 2hx + h^2) + k \\ &= ax^2 - \underbrace{2ahx}_b + \underbrace{ah^2 + k}_c \end{aligned}$$

$$\Rightarrow \frac{b}{-2a} = \frac{-2ah}{-2a} \Rightarrow \boxed{h = \frac{-b}{2a}}$$

$$\begin{aligned} \Rightarrow c &= ah^2 + k \Rightarrow k = c - ah^2 \\ &= c - a \left(\frac{-b}{2a} \right)^2 \\ &= c - \frac{ab^2}{4a^2} \end{aligned}$$

$$\boxed{k = c - \frac{b^2}{4a}}$$

Example. Find the vertex of $f(x) = 2x^2 - 6x + 7$ by

- computing h and k , and
- rewriting the quadratic in standard form by completing the square.

Identify if the vertex is the maximum or minimum value.

$$h = -\frac{b}{2a} = -\frac{-6}{2(2)} = \boxed{\frac{3}{2}} \quad k = c - \frac{b^2}{4a} = 7 - \frac{(-6)^2}{4(2)} = 7 - \frac{36}{8} = \boxed{\frac{5}{2}}$$

- Factor “ a ” from the first two terms

$$= 2(x^2 - 3x) + 7$$

- Add and subtract h^2

$$= 2\left(x^2 - \underbrace{3x}_{\frac{b}{a}} + \underbrace{\left(\frac{3}{2}\right)^2}_{+\left(\frac{b}{2a}\right)^2} - \underbrace{\left(\frac{3}{2}\right)^2}_{-\left(\frac{b}{2a}\right)^2}\right) + 7$$

- Bring $-h^2$ out

$$= 2\left(x^2 - 3x + \left(\frac{3}{2}\right)^2\right) - 2\left(\frac{3}{2}\right)^2 + 7$$

- Simplify

$$= 2\left(x - \frac{3}{2}\right)^2 - 2\left(\frac{9}{4}\right) + 7$$

$$= 2\left(x - \frac{3}{2}\right)^2 - \frac{9}{2} + 7\left(\frac{2}{2}\right)$$

$$= 2\left(x - \frac{3}{2}\right)^2 + \frac{5}{2}$$

Vertex at $\left(\frac{3}{2}, \frac{5}{2}\right)$

Minimum since $a=2 > 0$



Example. Find the vertex of $g(x) = 13 - 3x^2 + 4x$ by

- computing h and k , and
- rewriting the quadratic in standard form by completing the square.

Identify if the vertex is the maximum or minimum value.

$$h = -\frac{b}{2a} = -\frac{4}{2(-3)} = \boxed{\frac{2}{3}} \quad k = g(h) = c - \frac{b^2}{4a} = 13 - \frac{4^2}{4(-3)} = 13 + \frac{4}{3} = \boxed{\frac{43}{3}}$$

- Factor “ a ” from the first two terms

$$\begin{aligned} &= -3x^2 + 4x + 13 \\ &= -3\left(x^2 - \frac{4}{3}x\right) + 13 \end{aligned}$$

- Add and subtract h^2

$$= -3\left(x^2 - \frac{4}{3}x + \left(\frac{4}{2 \cdot 3}\right)^2 - \left(\frac{4}{2 \cdot 3}\right)^2\right) + 13$$

- Bring $-h^2$ out

$$= -3\left(x^2 - \frac{4}{3}x + \left(\frac{2}{3}\right)^2\right) - (-3)\left(\frac{2}{3}\right)^2 + 13$$

- Simplify

$$= -3\left(x - \frac{2}{3}\right)^2 + \frac{4}{3} + 13\left(\frac{3}{3}\right)$$

$$= -3\left(x - \frac{2}{3}\right)^2 + \frac{43}{3}$$

vertex at $\left(\frac{2}{3}, \frac{43}{3}\right)$

Maximum since $a = -3 < 0$

Example. For the following quadratic functions, find

- the x and y -intercepts,
- the vertex, and
- the domain and range.

x-intercepts: $\left(\frac{-1+\sqrt{13}}{3}, 0\right) \quad \left(\frac{-1-\sqrt{13}}{3}, 0\right)$

$$f(x) = \underbrace{3x^2}_a + \underbrace{2x}_b - \underbrace{4}_c$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-2 \pm \sqrt{(2)^2 - 4(3)(-4)}}{2(3)} = \frac{-2 \pm \sqrt{4 + 48}}{6} = \frac{-2 \pm 2\sqrt{13}}{6} = \frac{-1 \pm \sqrt{13}}{3}$$

y-intercept: $f(0) = -4 \rightarrow \boxed{(0, -4)}$

vertex: $\left(-\frac{1}{3}, -\frac{13}{3}\right)$ $f\left(-\frac{1}{3}\right) = 3\left(-\frac{1}{3}\right)^2 + 2\left(-\frac{1}{3}\right) - 4 = -\frac{13}{3}$

Domain: $(-\infty, \infty)$

Range: $\left[-\frac{13}{3}, \infty\right)$ $a = 3 > 0 \Rightarrow \text{min}$

$$g(x) = -(x+2)^2 - 1$$

x-intercepts: $-(x+2)^2 - 1 = 0$
 $-1 = (x+2)^2$
 $\pm\sqrt{-1} = x+2$
 $-2 \pm i = x$
 $\boxed{(-2-i, 0)} \quad \boxed{(-2+i, 0)}$

y-intercept: $g(0) = -(0+2)^2 - 1$
 $= -4 - 1$
 $= -5 \rightarrow \boxed{(0, -5)}$

vertex: $(h, k) = (-2, -1)$

Domain: $(-\infty, \infty)$

Range: $(-\infty, -1]$ $a = -1 < 0 \Rightarrow \text{max}$

Solving quadratic equations via factoring:

Example. Solve $2x^2 + x = 3x + 12$

1. Rewrite the equation in the general form: $2x^2 - 2x - 12 = 0$
2. Rewrite bx using factors of ac : $2x^2 - 6x + 4x - 12 = 0$
3. Factor out like terms: $2x(x - 3) + 4(x - 3) = 0$
4. Factor by grouping: $(x - 3)(2x + 4) = 0$
5. Solve for the roots: $x = 3$ and $x = -2$

Example. Solve the following for x via factoring:

$$(x + 3)(x - 1) = 5$$

$$x(x - 1) + 3(x - 1) = 5$$

$$x^2 - x + 3x - 3 = 5$$

$$x^2 + 2x - 8 = 0$$

Sign of larger number

Factors have diff signs

$$(x + 4)(x - 2) = 0$$

Order doesn't matter!

$$x(x + 4) - 2(x + 4) = 0$$

These should match!

$$(x - 2)(x + 4) = 0 \Rightarrow \begin{cases} x = 2 \\ x = -4 \end{cases}$$

$$(-1)(-4x^2 + 8x - 3) = 0$$

$$4x^2 - 8x + 3 = 0$$

Sign of larger number

Factors have same signs

$$4 \cdot 3 = 1 \cdot 12$$

$$\begin{matrix} 2 \cdot 6 \\ 3 \cdot 4 \end{matrix}$$

$$4x^2 - 2x - 6x + 3 = 0$$

$$2x(2x - 1) - 3(2x - 1) = 0$$

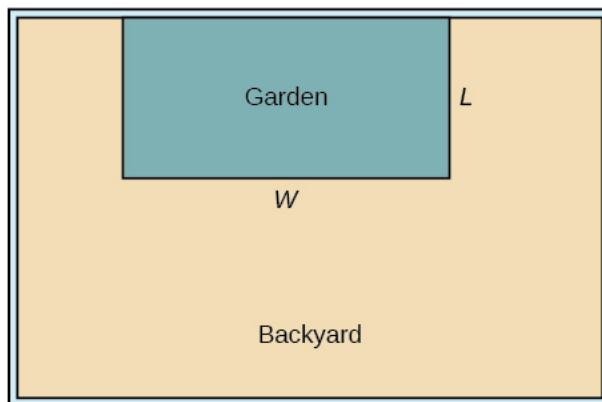
$$(2x - 3)(2x - 1) = 0$$

$$\Rightarrow \begin{cases} x = 3/2 \\ x = 1/2 \end{cases}$$

Example. A backyard farmer wants to enclose a rectangular space for a new garden within her fenced backyard. She has purchased 80 feet of wire fencing to enclose three sides, and she will use a section of the backyard fence as the fourth side (see below). What dimensions should she use to maximize the enclosed area?

$$A = W \cdot L$$

$$W + 2L = 80 \rightarrow W = 80 - 2L$$



$$\rightarrow A = (80 - 2L)L = 80L - 2L^2 = \underbrace{-2L^2}_a + \underbrace{80L}_b + \underbrace{0}_c$$

Since $a = -2 < 0$, the area is maximized at the vertex (h, k)

$$h = -\frac{b}{2a} = -\frac{80}{2(-2)} = 20 \text{ ft}$$

$$k = A(20) = -2(\textcolor{red}{20})^2 + 80(\textcolor{red}{20}) = 800 \text{ ft}^2$$